Chapter 9 Why Student's t appears

Detailed notes pp. 1–4

Lecture slides PDF pp. 1–18

The testing framework is unchanged: estimate minus null value, divided by its estimated SE. What changes is the reference distribution when population variability must be estimated.

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}, \qquad T = \frac{\bar{X} - \mu}{S/\sqrt{n}}.$$

For an independent random sample from a normal population, the first quantity is standard normal when σ is known, while the second has an exact t_{n-1} distribution when σ is unknown. Both numerator and denominator of T vary across samples.

Degrees of freedom and heavier tails

$$S^2 = \frac{\sum(X_i - \bar{X})^2}{n-1}.$$

The residuals sum to zero after estimating one mean, leaving $n-1$ degrees of freedom. Student's t curves are symmetric about zero with heavier tails than a standard normal curve; the extra tail probability reflects uncertainty in the estimated SD. As degrees of freedom increase, the curves approach the normal curve.

Read the upper-tail notation carefully

$$P(T_\nu > t_{\nu,q}) = q, \qquad \nu = df.$$

A two-sided confidence level $1-\alpha$ uses $t_{\nu,\alpha/2}$. A right-sided level-α test uses $t_{\nu,\alpha}$; a left-sided test uses its negative.

df	Upper 5%	Upper 2.5%
5	2.015	2.571
9	1.833	2.262
16	1.746	2.120
Normal limit	1.645	1.960

The assumption is not created by the sample size

Small n alone does not justify t. Exact results need the stated normal-sampling assumptions. Approximate normality gives approximate inference; strong skew, outliers or dependence can be serious at small sample sizes. The exact one-sample t result also holds for large normal samples.

Keep in mind. SD describes individuals. Estimated SE, $s/\sqrt{n}$, describes the sample mean's variability. Do not substitute one for the other.

Chapter 9 One population mean

Detailed notes pp. 5–9

Lecture slides PDF pp. 19–33

Conditions and target

Use an independent random sample from a normal population with unknown SD. The target is its mean μ; observed summaries are $n, \bar{x}, s$. Set $df = n - 1$ and estimated SE $= s/\sqrt{n}$.

Confidence interval

$$\boxed{CI_{1-\alpha}(\mu): \quad \bar{x} \pm t_{n-1,\alpha/2}\frac{s}{\sqrt{n}}.}$$

Interpret confidence as repeated-sample coverage of the fixed population mean. The interval does not contain a stated percentage of individual observations.

Hypothesis test

For $H_0 : \mu = \mu_0$, calculate

$$\boxed{t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad T \sim t_{n-1} \text{ under } H_0.}$$

Let F_ν denote the CDF of a t variable with ν degrees of freedom:

Alternative	p-value	Rejection rule
$\mu > \mu_0$	$1 - F_\nu(t_{\text{obs}})$	$t_{\text{obs}} > t_{\nu,\alpha}$
$\mu < \mu_0$	$F_\nu(t_{\text{obs}})$	$t_{\text{obs}} < -t_{\nu,\alpha}$
$\mu \neq \mu_0$	$2[1 - F_\nu(\lvert t_{\text{obs}}\rvert)]$	$\lvert t_{\text{obs}}\rvert > t_{\nu,\alpha/2}$

The tail logic is the same as Chapter 8; the reference is t, not Φ.

Connect test and interval

For the matching two-sided t test and CI, reject $H_0 : \mu = \mu_0$ at level α exactly when the $1 - \alpha$ interval excludes μ_0, apart from boundary conventions. A two-sided 95% CI does not directly implement every one-sided 5% decision.

Revisit the slide examples

The diamond example on p. 6 demonstrates a CI from raw data. The paint-coverage example on p. 9 demonstrates a two-sided test. Reconstruct parameter, SE, *df*, tail and interpretation before checking the worked calculation.

Keep in mind. A table can give a p-value range rather than a precise decimal. Do not manufacture extra precision when only bounding table entries are available.

Chapter 9 Two independent means

Detailed notes pp. 10–14

Lecture slides PDF pp. 34–55

The pooled model has an extra assumption

Two independent random samples come from normal populations with **a common unknown population variance** σ^2. The sample SDs need not be numerically equal. The course uses $1/3 < s_1^2/s_2^2 < 3$ as a screening rule of thumb (slide p. 46; detailed p. 11). This is not proof that the population variances are equal. Define $\Delta = \mu_1 - \mu_2$ with a fixed group order.

$$\boxed{s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$$

The pooled variance combines squared deviations, weighted by their degrees of freedom. It is not the simple average of SDs.

$$\widehat{\mathrm{SE}}(\bar{x}_1 - \bar{x}_2) = s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}, \qquad df = n_1+n_2-2.$$

Interval and test

$$CI_{1-\alpha}(\Delta): \quad (\bar{x}_1-\bar{x}_2) \pm t_{n_1+n_2-2,\alpha/2}\, s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}.$$

For $H_0 : \Delta = \Delta_0$,

$$\boxed{t_{\text{obs}} = \frac{(\bar{x}_1-\bar{x}_2)-\Delta_0}{s_p\sqrt{1/n_1+1/n_2}}.}$$

Use the t tails from p. 2 with $df = n_1+n_2-2$. Under the stated normal and equal-variance assumptions, this reference is exact.

What pooling means here

Detailed notes pp. 11: pooled variance and its assumptions

This pools estimates of a *common variance*. Chapter 8's two-proportion test pools a *common success probability under the null*. They are different models; sharing the word "pooled" does not make their formulas interchangeable.

Check design and assumptions before computing

- Different independent individuals in the two groups justify an independent-samples structure.
- Equal sample sizes do not imply pairing; unequal sample sizes do not prevent independent comparison.
- If equal population variances are not justified, do not silently use the pooled method. An unequal-variance procedure is a different method, beyond this deck's pooled formula.

Keep in mind. The online/classroom example on detailed pp. 12–14 corrects source arithmetic while retaining the original data. Use its checked calculations when reviewing the slide example.

Chapter 9 Paired differences and method choice

Detailed notes pp. 15–17

Lecture slides PDF pp. 56–62

When the same individuals are measured twice, or observations are explicitly matched, analyze within-pair differences. Choose an order and define $D_i = X_i - Y_i$; the target is μ_D.

Reduce to one-sample t inference

For n independent pairs whose differences are normally distributed,

$$\bar{d} = \frac{1}{n}\sum d_i, \qquad s_D^2 = \frac{\sum(d_i - \bar{d})^2}{n-1}, \qquad df = n - 1.$$

$$\boxed{CI_{1-\alpha}(\mu_D): \quad \bar{d} \pm t_{n-1,\alpha/2}\frac{s_D}{\sqrt{n}}}$$

$$\boxed{t_{\text{obs}} = \frac{\bar{d} - d_0}{s_D/\sqrt{n}} \quad \text{for } H_0 : \mu_D = d_0.}$$

The required normality concerns the *differences*, and the required independence concerns different pairs. Within-pair dependence is expected and is retained through s_D. Reversing subtraction reverses signs and interval endpoints' order after negation; reverse the alternative consistently too.

Select by study design

Design	Course small-sample normal method
One sample; one population mean	One-sample t, $df = n - 1$.
Separate independent groups; common variance justified	Pooled two-sample t, $df = n_1 + n_2 - 2$.
Repeated or matched observations	One-sample t on differences, $df =$ number of pairs -1.

The final explanation matters

State the parameter, assumptions, estimate, SE, degrees of freedom, reference, tail and conclusion. Fail-to-reject does not prove equality. An interval for a mean difference is not an interval for every individual's change. The paired tire example on detailed p. 16 shows the workflow.

Keep in mind. Use existing released course examples for review. Future Chapter 9 practice and review questions are selected separately; this guide does not announce a new practice set or predict the exam's questions.